Programming Robots 💻

Junior Robotics Badge — Girls Build Bots

Presented by your robotics mentors!

Quick review: what's a program?

A program = a list of instructions a computer follows **exactly**.

But real programs have superpowers: - **Loops** — repeat steps automatically 🔁 - **Events** — react when something happens ⚡ - **Conditions** — make decisions 🤔

Today you'll use all three!

Loops: Don’t Repeat Yourself! 🔁

Dance routine WITHOUT a loop: > clap, clap, spin, jump, clap, clap, spin, jump, clap, clap, spin, jump

Dance routine WITH a loop: > **repeat 3 times:** [clap ×2, spin, jump]

Same dance. Way less writing. **Let’s perform it!** 💃

Events: When THIS happens, do THAT

Events make programs interactive:

Event	Action
arrow key pressed	move the player
two sprites touch	collect the item
timer hits zero	game over

Robots use events too: *when bump sensor pressed → turn around*

Build: Battery Chase Game 🎮

In MakeCode Arcade, step by step:

1. **Sprite + events** — player moves with arrow keys
2. **Second sprite** — a battery to collect 🔋
3. **Loop** — battery respawns *forever*
4. **Condition** — IF score = 5 → “YOU WIN!”
5. **Challenge** — add an enemy that chases you! 👾

Finished early? Become a **deputy mentor** and help a neighbor!

Debugging like a pro 🐛

When your game doesn’t work: 1. **Read** your blocks out loud 2. **Predict** what should happen 3. **Compare** to what actually happens 4. **Change ONE thing** and test again

Never change five things at once — you won’t know which one fixed it!

Your game IS a robot brain 

Your game	A real robot
Player sprite	The robot
Arrow keys	Sensor inputs
Enemy touching you	Obstacle detected
“YOU WIN” condition	Mission complete

Think like a roboticist 

Write **pseudocode** (a program in plain English) for a vacuum robot:

```
forever:
    move forward
    if bump sensor pressed:
        turn right
```

Your turn: write pseudocode for a robot that waters plants! 

You did it! 🎉

Today you: - ✅ Used loops, events, and conditions - ✅ Built a complete game - ✅ Debugged like a professional - ✅ Wrote pseudocode for a real robot task

Keep going at home! 🏠

- **arcade.makecode.com** — keep building your game (free)
- **scratch.mit.edu** — make games and animations
- **code.org** — free coding courses
- **micro:bit** (microbit.org) — a $20 pocket computer you can program
- Ask about **FIRST LEGO League** teams near you!